

Urshita Pal

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EDUCATION

University of Michigan, USA

Ph.D., Mathematics, 2021-Present.

Advisor: Jenny Wilson

University of Michigan, USA

M.S., Mathematics, 2021-23.

Chennai Mathematical Institute, India

B.Sc., Mathematics & Computer Science, 2018-21.

RESEARCH INTERESTS

(Co)homology of Arithmetic Groups, (Co)homological stability phenomena

AWARDS AND FELLOWSHIPS

Edwin Wilkinson Miller award

Summer 2026

Rackham Predoctoral Fellowship

Fall 2025 - Summer 2026

Rackham One-Term Dissertation Fellowship

January 2025 - April 2025

Gold Medal of Excellence, BSc Math & Computer Science

Chennai Mathematical Institute, 2021

Highest GPA at time of graduation

Tuition Fee Waiver

Chennai Mathematical Institute, Aug 2018 - April 2021

Bronze Medal, European Girls Math Olympiad

Held in Florence, Italy, 2018

International Math Olympiad Training Camp, Mumbai, India

Selected for Participation, 2017 & 2018

RESEARCH PAPERS

A projective resolution of the symplectic Steinberg module

Arxiv preprint Arxiv:2605.06499

The top cohomology of principal congruence subgroups of special linear groups over Euclidean number rings

Arxiv preprint Arxiv:2605.05087

Representation stability in the (co)homology of vertical configuration spaces

with D Baron, C Wang, J Wilson, and C Yang

ArXiv preprint ArXiv:2412.01128

TEACHING

University of Michigan

Instructor, Math 105 (Data, Graphs & Functions), Fall 2024

Instructor, Math 115 (Calculus 1), Fall 2023

Instructor, Math 115 (Calculus 1), Winter 2023

Instructor, Math 115 (Calculus 1), Fall 2022

Instructor, Math 115 (Calculus 1), Winter 2022

Instructor, Math 115 (Calculus 1), Fall 2021

Chennai Mathematical Institute

Teaching Assistant, NPTEL Rings & Fields, Jan-March 2021

Teaching Assistant, Topology, Spring 2021

Teaching Assistant, Complex Analysis, Spring 2021

Teaching Assistant, Analysis 3, Fall 2020

Teaching Assistant, Probability Theory, Spring 2020

TALKS

Invited Talks

- *The top cohomology of principal congruence subgroups of $SL_n R$ and $Sp_{2n} R$*
Binghamton University, April 2026

- *The top cohomology of principal congruence subgroups of $SL_n R$ and $Sp_{2n} R$*
Spring Redbud Topology Conference, University of Oklahoma, April 2026

- *Representation Stability in the (Co)homology of Vertical Configuration Spaces*
Special Session at AMS Spring Sectional, Boston College, March 2026

- *The generalized Lee–Szczarba conjecture on the cohomology of principal congruence subgroups*
UC San Diego, February 2026

- *A Resolution of the Symplectic Steinberg Module*
Purdue University, October 2025

- *Steinberg Modules and Hopf Algebras* (expository)
University of Pennsylvania, March 2025

- *Steinberg Modules and $H^*(SL_n \mathbb{Z}; \mathbb{Q})$* (expository)
Scissors Congruences, Algebraic K-Theory and Steinberg Modules, July 2024
(Held at the American Institute of Mathematics, Pasadena)

SERVICE AND
MENTORSHIP

Summer Workshops

- *High Connectivity of Arc Complexes* (expository)
Talbot Workshop on Homological Stability, May 2025

- *Connectivity and Cohomology* (5 min lightning talk)
Young Geometric Group Theory, Bristol, April 2024

University of Michigan REU, Summer 2024

- *Graduate Student Mentor for the REU hosted by Jenny Wilson*

Lab of Geometry at Michigan

- *Served on the Admissions Committee in Fall '23 & Winter '24*

Student Dynamics/Geometry/Topology Seminar

- *Co-organiser 2022-23, Organiser 2023-24*

AWM Mentoring Programs

- *Mentor in the Mentor-Mentee program for the Academic Year 2022-23*
- *Participated in the 'Vertical Mentoring Program' in Winter 2024*

Michigan Directed Reading Program

- *Mentor, Winter 2023, Winter 2025, and Fall 2025*

Michigan Math Club

- *Fagnano's Problem and Reflecting Triangles, Jan 2023*

Michigan Math Circle

- *Tiling With Dominoes, Feb 2022*

CONFERENCES
ATTENDED

- Spring Redbud Topology Conference, University of Oklahoma, April 2026
- Perspectives on the Cohomology of $GL_n\mathbb{Z}$, Muenster, Germany, October 2025
- Scissors Congruences and Algebraic K-Theory, UPenn, July 2025
- Discrete Groups in Topology & Algebraic Geometry, Notre Dame, June 2025
- Polylogarithms, Homology of linear groups, & Steinberg Modules, June 2025
Held at the SwissMAP Research Station, Les Diablerets, Switzerland
- Talbot Workshop on Homological Stability, Cassapolis MI, May 2025
- Scissors Congruences, Algebraic K-Theory and Steinberg Modules, July 2024
Organised through the American Institute of Mathematics, Pasadena
- Young Geometric Group Theory XII, Bristol, April 2024
- Stability in Topology, Arithmetic and Representation Theory, Purdue, 2023
(Attended Virtually)

- Nearly Carbon Neutral Geometric Topology Conference, June 2023
(Attended Topic Group: Profinite and Residual Methods in Geometric Group Theory)

- Nearly Carbon Neutral Geometric Topology Conference, September 2022
(Attended Topic Group: Group Actions on Hyperbolic Spaces)